Field Hours

CSCE 658 Randomized Algorithms

Original notes for this site. Same topics as the catalog course. Not a reprint of another university's lectures, and not academic credit.

Lecture 1 · 90 minutes

Indicators and linearity

You analyze a randomized algorithm by turning events into numbers. An indicator is 1 when its event happens and 0 otherwise, and the expectation of that indicator equals the probability of the event. Toss 8 fair coins. The number of heads is the sum of 8 indicators, each with expectation 1/2, so the expected number of heads is 4. Dependence among the coins does not break the sum.

You apply the same sum to records in a random permutation of 4 distinct keys. Let X_i be 1 when position i is larger than every key before it. Position 1 is always a record, so its expectation is 1. At position k the largest among the first k keys is equally likely to sit in any of those k places, so the expectation is 1/k. Adding 1 + 1/2 + 1/3 + 1/4 gives 25/12.

From an event to a sum of expectations.

Name the event -> Write a 0-1 indicator -> Take its expectation -> Add across events

Records among 4 keys

1. The permutation is 2, 4, 1, 3. Position 1 holds 2, a record. Position 2 holds 4, larger than 2, a record.

2. Position 3 holds 1, smaller than 4, so it is not a record. Position 4 holds 3, also smaller than 4.

3. This one permutation has 2 records. The expectation over all 24 permutations is 25/12, not the count on this sample.

Problem 1. A random permutation of 3 distinct keys. What is the expected number of left-to-right records?

Answer. Linearity gives 1 + 1/2 + 1/3 = 11/6. Averaging the 6 permutations by hand gives the same 11/6.

Problem 2. One fair coin. Let X be 1 on heads and Y be 1 on that same heads. What is E[X+Y]?

Answer. Each expectation is 1/2. Linearity ignores the dependence, so E[X+Y] = 1. The outcomes are 2 and 0, and their average is 1.

Lecture 2 · 90 minutes

Markov and Chebyshev tails

You reach for Markov when a random variable is nonnegative and you know only its mean. If the mean is m, the probability that the variable is at least t is at most m/t. Take a running time whose expectation is 10. Markov says the chance it runs for 40 or more is at most 10/40, which equals 1/4. The bound can be loose. It is available

before you know a variance.

You tighten the tail once a variance is in hand. Chebyshev says the probability of landing at least k standard deviations from the mean is at most 1/k squared. Suppose the mean is 20 and the variance is 16, so one standard deviation is 4. A deviation of 12 is 3 standard deviations, and the probability of such a deviation is at most 1/9. You paid for that bound by computing a second moment.

Two tail bounds, and what each one spends.

Markov

Needs a nonnegative variable

Uses only the mean

Mean 10 at 40 is 1/4

Chebyshev

Needs a variance

Counts standard deviations

3 deviations: at most 1/9

A running time with mean 10

1. You know only that T is nonnegative and E[T] = 10. Markov at threshold 50 gives probability at most 10/50 = 1/5.

2. You then learn the variance is 25, so the standard deviation is 5. The distance from 10 to 50 is 40, which is 8 standard deviations.

3. Chebyshev bounds that event by 1/8 squared = 1/64. The Markov number was 1/5. The variance is what improved the bound.

Problem 1. Nonnegative X has expectation 6. Use Markov to bound the probability that X is at least 24.

Answer. At most 6/24 = 1/4.

Problem 2. The mean is 10 and the variance is 4. Bound the probability that X is at least 6 away from the mean.

Answer. The standard deviation is 2, so a distance of 6 is 3 standard deviations. Chebyshev gives at most 1/9.

Lecture 3 · 90 minutes

Chernoff bounds for coin sums

You get a sharper tail when a sum is made of independent coin flips. A Chernoff bound builds the moment generating function of each flip, multiplies those functions, and applies Markov to the exponential. Take 100 flips of a fair coin, so the mean is 50. The chance of 75 or more heads, using the form exp(-m d squared / 3) with d = 1/2, is at most exp(-25/6), about 0.0155. Check independence before you quote it.

You size a repetition with that same bound. One trial of a test errs with probability 1/3, and you want a majority of independent trials to err with probability under 1/100. The error count has mean n/3. A threshold at half the trials is a relative excess of 1/2. At mean m = 60, hence n = 180, the bound exp(-m / 12) equals exp(-5), about 0.00674, under 1/100. Repetition, not a new test, is the mechanism.

How a Chernoff bound is assembled.

1. Start from each flip

2. Multiply the transforms

3. Apply Markov

4. Read the tail

100 fair coins

1. The mean is 50. You ask for the probability of at least 75 heads, a relative excess d = 1/2.

2. The bound exp(-m * d * d / 3) becomes exp(-50 * 0.25 / 3) = exp(-25/6), about 0.0155.

3. Chebyshev, with variance 25, only limits a deviation of 25 by 25/625 = 1/25. Chernoff is the tighter number here.

Problem 1. 200 fair coins. Use exp(-m d squared / 3) to bound the probability of at least 150 heads.

Answer. The mean m is 100 and the relative excess d is 1/2. The bound is exp(-100 * 0.25 / 3) = exp(-25/3), about 0.000240.

Problem 2. A test errs with probability 1/3. You run 180 independent trials and take a majority. Is the Chernoff failure bound under 1/100?

Answer. Yes. The error count has mean 60. Crossing half the trials is a relative excess of 1/2, and exp(-60 / 12) = exp(-5), about 0.00674, which is under 1/100.

Lecture 4 · 90 minutes

Las Vegas and Monte Carlo

You classify an algorithm by what the coins may spoil. A Las Vegas algorithm always returns a correct answer, and the running time is the random piece. Randomized quicksort on 8 distinct keys is Las Vegas: every pivot sequence sorts the array, while the usual bound 2 n ln n is about 33 comparisons. A Monte Carlo algorithm stops on a fixed budget and may return a wrong answer. Name the class before you discuss speed.

You shrink one-sided Monte Carlo error by repeating the test. Suppose a yes instance is accepted with probability at least 1/2, and a no instance is always rejected. Run the test 5 times and accept if any run accepts. A yes instance is missed on every run with probability at most 1/32. A no instance is never accepted. The time cost is a factor of 5, and the error on yes instances fell from 1/2 to 1/32.

What the coins are allowed to affect.

Las Vegas

Answer is always correct

Time is the random part

Quicksort on 8 keys

Monte Carlo

Time budget is fixed

Answer may be wrong

Repeat to shrink error

Five-fold one-sided amplification

1. One run accepts a yes instance with probability at least 1/2 and never accepts a no instance.

2. You run the test 5 times and accept if any run accepts. A no instance produces 5 rejections.

3. A yes instance misses every run with probability at most (1/2)^5 = 1/32. The time factor is 5.

Problem 1. Same one-sided test, repeated 7 times, accept if any run accepts. What is the miss probability on a yes instance?

Answer. (1/2)^7 = 1/128.

Problem 2. Is randomized quicksort on 8 keys Las Vegas or Monte Carlo, and what do the coins control?

Answer. Las Vegas. Every seed returns the sorted array. The coins choose pivots and affect the comparison count, whose usual bound is about 33, not the correctness.

Lecture 5 · 90 minutes

Universal hashing and chain length

You hash 4 keys into a table of 16 slots using a random function from a universal family. For any two fixed keys the probability they land in the same slot is at most 1/16. There are 6 unordered pairs, so the expected number of colliding pairs is at most 6/16, which is 3/8. Each pair contributes an indicator. Universality is the only property you used, and linearity sums the indicators even though pairs share keys.

You compare that pair count with chain length under a fully random hash. Each of 4 keys picks one of 16 slots uniformly and independently. The expected number of keys in one fixed slot is 4/16, which is 1/4. A search for a stored key scans that one chain. Pair counts and chain lengths summarize the same throws differently. You charge the summary that matches the operation you are timing.

Two summaries of 4 keys hashed into 16 slots.

4 keys into 16 slots

6 pairs, at most 3/8 collisions

One slot expects load 1/4

Pairs versus chains

1. A universal family puts any fixed pair in one slot with probability at most 1/16. Six pairs give expected collisions at most 6/16 = 3/8.

2. A fully random hash puts each key in a uniform slot. One fixed slot then has expected load 4/16 = 1/4.

3. One unlucky sample can still place 2 of the 4 keys together. The expectations do not forbid that sample.

Problem 1. 8 keys are hashed into 32 slots by a universal family. Upper-bound the expected number of colliding pairs.

Answer. There are 8*7/2 = 28 pairs. Each collides with probability at most 1/32, so the expectation is at most 28/32 = 7/8.

Problem 2. Fully random hashing of those same 8 keys into 32 slots. What is the expected load of one fixed slot?

Answer. 8/32 = 1/4.

Lecture 6 · 90 minutes

Contracting random edges for a min-cut

You search for a global min-cut by contracting random edges. While more than 2 vertices remain, you pick an edge uniformly, merge its endpoints, drop self-loops, and keep parallel edges. On n vertices the chance that one run preserves a fixed min-cut is at least 2/(n(n-1)). For n = 4 that lower bound is 2/12, which equals 1/6. A single run can destroy the cut. The quantity you compute is that failure probability.

You amplify the weak run by independent repetition. For n = 4 the failure probability of one run is at most 5/6. Two independent runs both fail with probability at most 25/36, and you keep the smaller cut you found. In general, C(n,2) times ln n runs push the failure probability under 1/n, because (1 - 1/C(n,2)) raised to that power is at most 1/n. The mechanism is repetition of a Monte Carlo procedure.

One contraction run, then a repetition.

1. Pick a uniform edge

2. Merge the endpoints

3. Stop at 2 vertices

4. Keep the smaller cut

Two runs on 4 vertices

1. For n = 4 one run succeeds with probability at least 2/(4*3) = 1/6, so failure is at most 5/6.

2. Two independent runs both fail with probability at most (5/6)^2 = 25/36.

3. You keep the smaller cut. The chance at least one run succeeded is at least 1 - 25/36 = 11/36.

Problem 1. n = 5. Lower-bound the success probability of one contraction run on a fixed min-cut.

Answer. 2/(5*4) = 1/10.

Problem 2. n = 4 and three independent runs, each failing with probability at most 5/6. Upper-bound the probability that all three fail.

Answer. (5/6)^3 = 125/216.

Lecture 7 · 90 minutes

Random cuts and conditional expectation

You cut a graph by placing each vertex on one side with a fair coin, independently. Any fixed edge crosses the cut with probability 1/2, because the second endpoint disagrees with the first half the time. On a graph with 10 edges, linearity makes the expected cut size 5. The optimum is at most 10, so this expectation is at least half the optimum. The coins are the whole algorithm. You do not list the partitions.

You remove the coins with the method of conditional expectation. Fix vertices one by one. When you place vertex 3, compute the expected cut with that vertex on side A and on side B, leaving later vertices random. The two numbers average to the expectation you already hold, so one of them is at least 5 on the 10-edge graph. Keep that side. The final cut is deterministic and at least 5.

Conditional expectation removes one vertex of coins at a time.

Compute both sides -> Keep the larger mean -> Fix the next vertex -> Finish at 5 or more

Derandomize a triangle

1. A triangle has 3 edges, so a fair random cut has expectation 3/2.

2. Fix A on the left. If B is left, AB is uncut and random C contributes expected cut 1, total 1. If B is right, AB is cut and C still contributes 1, total 2. Keep B on the right.

3. Either side for C then yields cut size 2, which is at least 3/2. The all-left assignment has cut 0 and is refused because the conditional expectation would drop.

Problem 1. A graph has 20 edges. What is the expected cut under fair independent side assignments, relative to the optimum?

Answer. The expected size is 10. The optimum is at most 20, so 10 is at least half the optimum.

Problem 2. The two conditional expectations for the next vertex are 7 and 5, and the current value is 6. What do you carry forward?

Answer. The options average to (7+5)/2 = 6, matching the current value. You keep the side with conditional expectation 7.

Lecture 8 · 90 minutes

What you can now prove about the coins

You can now pick a paradigm and finish the calculation. For a sum of indicators you write the expectation by linearity, even when the events share coins, as with records in 4 keys and with 10 cut edges. For a tail you choose Markov, Chebyshev, or Chernoff from what you know: a mean, a variance, or independent bits. For a Monte

Carlo run that succeeds with probability only 1/6, you turn to repetition and multiply the failure bound.

You can also say what the coins are allowed to spoil. Quicksort on 8 keys returns a sorted array on every seed, so it is Las Vegas and the bound near 33 is a time bound. One random cut of a 10-edge graph can miss the optimum, while the expectation 5 is a guarantee about the average sample. Your last distinction is the promise attached to the number: correctness every time, an expectation, or a tail.

The toolkit, from the coins to the promise.

Indicators and linearity

Markov, Chebyshev, or Chernoff

Las Vegas or Monte Carlo

Repeat a weak success probability

Repeat a 4-vertex min-cut run

1. On 4 vertices one contraction fails with probability at most 5/6.

2. Two independent runs therefore fail together with probability at most 25/36. That product is the amplification step.

3. The repeated procedure succeeds with probability at least 11/36. You multiplied failure bounds rather than re-deriving the contraction.

Problem 1. Combine the min-cut success bound with amplification. One run on 4 vertices succeeds with probability at least 1/6. What is an upper bound on both of 2 independent runs failing?

Answer. Failure on one run is at most 5/6. Independence lets you multiply, so both fail with probability at most 25/36. At least one success then has probability at least 11/36.

Problem 2. Combine the random cut with Markov. A fair cut of 10 edges has expected size 5. Let Y = 10 minus the cut size. Upper-bound the probability that the cut is empty.

Answer. Y is nonnegative and E[Y] = 5. An empty cut means Y = 10. Markov gives P(Y >= 10) at most 5/10 = 1/2.